25900357 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCECT204 - DIGITAL FUNDAMENTALS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. Convert $(A2)_{16}$ to decimal. [U] / [CO1]

   a). 160 b). 161 c). 162 d). 163

2. A sequential circuit requires memory with no undefined state and minimal design complexity. Which flip-flop would you recommend? [U] / [CO2]

   a). RS b). JK c). T d). D

3. State the role of a Programmable Interrupt Controller (PIC). [U] / [CO3]

   a). To manage memory allocation

   b). To handle interrupt requests and prioritize them

   c). To execute system calls

   d). To perform context switching

4. Identify the output of an XNOR gate when both inputs are the same. [U] / [CO1]

   a). 1 b). 0

c). both A & B d). none

5. According to DeMorgan's theorems, the complement of a sum of variables is equal to [U] / [CO2]

   a). The complement of the sum

   b). The sum of the complements

   c). The product of the complements

   d). None of the above

6. Choose the primary advantage of microprogram control. [K] / [CO3]

   a). Faster execution

   b). More complex design

   c). Flexibility in instruction set design

   d). Less memory usage

7. Simplify the expression (A+B)(A+C): [U] / [CO1]

   a). A + BC b). A + B + C c). A + BC + AC d). A + B

8. Which addressing mode does the instruction ADD R1,#10 use? [U] / [CO2]

   a). Immediate b). Register c). Direct d). Idirect

9. Identify the correct description of Direct Memory Access (DMA). [K] / [CO3]

   a). CPU accesses memory directly

   b). Peripherals access memory directly without CPU

   c). Memory bypasses I/O devices

   d). CPU and memory transfer simultaneously

10. Calculate the hexadecimal representation of the binary number 11010111. [A] / [CO2]

    a). D7 b). E6 c). B7 d). F7

[1x10 = 10]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **5** marks

11.	Discuss the alternative method to find the 2's complement method with an example.	[U] / [CO1]
12.	A user saves a document in a computer, but when power is turned off, the data disappears. Apply the concept of memory types to identify which type of memory was used and suggest a suitable alternative.	[U] / [CO2]
13.	Identify and explain the two processor registers that is used to transfer data between memory and processor .	[U] / [CO3]
14.	Illustarte the working of a half adder circuit and truth table.	[K] / [CO2]
15.	Explain the working of PIPO shift register	[U] / [CO2]
16.	List the advantages of flash memory.	[K] / [CO3]

[5x4 = 20]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

17.	Discuss advantages, disadvantages and applications of BCD representation in computers.	[U] / [CO1]
18.	Design a circuit to implement a 3 to 8 decoder considering active high logic.	[U] / [CO2]
19.	List the mapping techniques used in cache memory.	[K] / [CO3]

[10x2 = 20]